

YUGESH BHOGE

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PUBLICATION

- 1. Distinguishing Kilonovae from Binary Neutron Star and Neutron Star-Black Hole Mergers** March 13, 2026
Ish Gupta, **Yugesh Bhoge**, Rahul Kashyap, Mukul Bhattacharya
◦ [arXiv:2603.13484](#) (In review at ApJL, AAS75361)

RESEARCH EXPERIENCE

- 1. Bayesian formalism for distinguishing kilonova from NSBH and BNS mergers** (* Manuscript in preparation)
Jul 2024 - March 2026
Guided by Prof Rahul Kashyap IIT Bombay
◦ Performing parameter estimation upon intrinsic CBC parameters solely from lightcurve involves converting lightcurve properties, such as ejecta mass and velocity, into individual component masses and tidal deformability. This will help in **determining CBC parameters solely based on EM observation**, and also enable breaking the degeneracy between the merger types.
- 2. Constraining Equation of State of neutron star using neutron star-black hole mergers *** Feb 2024 - Aug 2025
In collaboration with Mr Vasudev Dubey, Prof Rahul Kashyap and Dr Ish Gupta IIT Bombay
◦ Built a Bayesian inference pipeline based on Bilby with **IMRPhenomNSBH** and **IMRPhenomNRTidalV2** waveform models to recover tidal deformability from simulated NSBH merger signals, achieving accurate retrieval of chirp mass, mass ratio and effective tidal deformability ($\tilde{\Lambda}$).
- 3. Rayleigh-Taylor Instability in Stellar Interiors**  Jan 2025 - Apr 2025
Guided by Prof Rahul Kashyap IIT Bombay
◦ Derived a **general MHD dispersion relation** for the Rayleigh Taylor instability with stability map plots in stellar interiors, predicted critical/fastest growing wavelengths and confirmed the theory with **2D finite volume simulations**. Our results confirm the **stabilising influence of magnetic fields**, especially for **shorter wavelengths**, and highlight conditions under which RTI is suppressed.
- 4. Design and Fabrication of Suspension System for All-Terrain Vehicle** Mar 2019 - Apr 2020
Guided by Prof Sunil Dambare Pune University
◦ Designed, analysed, fabricated and tested the suspension system for an Off-road racing vehicle with the use of different software. Minimised the camber gain on wheels to **0.0098deg per mm of wheel travel**, also maintained the ride frequency to **1 - 1.8 Hz (front) and 1.1 - 2.0Hz (rear)** to ensure comparatively better ride quality, verified with actual experimentation, found to be within **1.6%**
- 5. Study on Aerospike Engine and its Performance Comparison with Conventional Bell-Cone type Nozzle in Single Stage To Orbit Flight**  Mar 2019 - Nov 2019
Guided by Prof Sunil Dambare Pune University
◦ Demonstrated the effectiveness of the aerospike nozzle over conventional bell cone type nozzle (both modelled and tested in computational environment using CAD and CFD) by comparing their properties like **PMF**, which is decreased by **7.00%**, and **thrust coefficient**, which is increased by **26.66%**. This can make a single stage to orbit possible, also reducing the amount of fuel needed and hence can increase payload capacity.

PROJECTS

- 1. Blindspots in omnidirectional Interferometric Gravitational Waves detectors** Oct 2024 - Dec 2024
Tools: PyCBC 
◦ Investigated the directional sensitivity and blind spots for gravitational wave detectors (**LIGO, Virgo and KAGRA**) using **PyCBC**, computed the antenna pattern functions and proposed mitigation strategies through multi-detector networks to improve detection sensitivity
- 2. Demonstration of chaos by period doubling route using Malkus waterwheel** Aug 2024 - Oct 2024
Tools: Python, ImageJ 
◦ Built the Malkus waterwheel alongside its mathematical model based on **Lorenz system equation, limit cycle, pitchfork**, and **subcritical Hopf bifurcation** and demonstrated its chaotic behaviour under steady flow using **self-written code in Python** and used **ImageJ** software for the image analysis and tracing purpose
- 3. Transforming Alphabets into Touch: A Braille Language Project** Sep 2023 - Dec 2023
Tools: Arduino UNO, Arduino IDE, C++ 
◦ Mapped all **26** letters of Braille language into **6 segment display**, also included null characters as a significant time delay pause so that visually and hearing impaired persons can distinguish between the words in a sentence and sentences from paragraphs. It uses a 6-segment haptic feedback display using 6 servo motors controlled using **Arduino UNO**
- 4. The Eco Coach** May 2018 - Dec 2019
Design Project
◦ Design project done for the Indian Railways to mitigate the problem of littering of plastic on railway tracks, built an integrated shredder and storage assembly for storing plastic waste inside the coach

EDUCATION

- **Indian Institute of Technology, Bombay** Jul 2023 - May 2025
MSc Physics
◦ CPI: 7.54/10.00
Mumbai, India
- **Savitribai Phule Pune University** Aug 2016 - Apr 2020
BTech Mechanical Engineering
◦ CGPA: 8.85/10.00
Pune, India

PROFESSIONAL TEACHING AND RESEARCH EXPERIENCE

- **Freelance Educator & Independent Professional** Dec 2020 – Jul 2023
◦ Worked in a corporate role immediately following graduation (Dec 2020 – Apr 2021). Transitioned to freelance, provided private tutoring in physics and mathematics to high school and undergraduate students. Concurrently prepared for JAM 2023, qualifying with 99.45th percentile.
- **Independent Researcher & Gen AI expert** May 2025 – Present
Multi-messenger Astrophysics, IIT Bombay Remote collaboration
◦ Continuing kilonova population analysis; paper submitted to ApJL (arXiv:2603.13484, March 2026), currently under review. Developing NSBH EoS constraints pipeline (manuscript in preparation). Presented at YAM 2026; active LIGO Scientific Collaboration member. Also working on improving AI models to predict experimental outcomes.

TALKS AND WORKSHOPS

- **The Young Astronomers Meeting 2026** Mar 2026
Hybrid CAMK, Poland
◦ Attended 3 days meeting where young and veteran collaborator over the world shared their work. Presented my recently published work.
- **GW - Detector Characterization Workshop** Dec 2025
LISC IUCAA, Pune, India
◦ Hands-on training including noise subtraction, glitch characterisation, spectral analysis, and machine learning applications in LIGO data analysis.

HONORS AND AWARDS

- **Joint Admission for Masters** May 2023
◦ Secured 99.45th percentile in nationwide entrance examination for masters study
- **INSPIRE Scholarship** May 2016
Department of Science & Technology, Government of India
◦ Awarded to the top 1 % of the national Class 12 cohort for academic excellence

PROFESSIONAL MEMBERSHIPS

- **LIGO, LSC and LISC**, Membership ID: yugeshjivaram.bhoge@ligo.org Feb 2025 - Present

SKILLS

- **Computational Astrophysics:** Bilby, LALSuite, PyCBC, MOSFiT, NumPy, SciPy, Astropy
- **Simulation & Engineering:** FVM, CFD, CAD, Arduino IDE, LTSpice, ImageJ
- **Mathematical & Statistical Tools:** MATHEMATICA, L^AT_EX, OriginLabs, MATLAB, Scilab
- **Programming & HPC:** Python, C++, Bash, HTML, HTCondor, Cuda, Cray

VOLUNTEER AND OUTREACH EXPERIENCE

- **PE ROTA volunteer** Aug 2025
LIGO Scientific Collaboration
◦ Volunteered to launch 1 manual PE run for vanilla BBH event in PE ROTA shift
- **Scientific summary translator** Jul 2025
LIGO Scientific Collaboration
◦ Translated an exceptional event's (GW231123) science summary into regional languages (Marathi)
- **LIGO India Outreach Volunteer** Dec 2024
LIGO INDIA 
◦ Volunteered at TechConnect (Techfest) with eight peers to raise awareness about gravitational waves and dispel misconceptions around the detectors by organising engaging educational activities, like the "Black Hole Ping Pong" game and an interactive "Stretch and Squash" selfie tool, to simplify complex concepts for diverse audiences

CERTIFICATIONS

- **International Centre for Theoretical Science, Bangalore:** Fascinating worlds of flows Jul 2023
- **Dassault's Systemes:** Certified SolidWorks Professional Dec 2020
- **Coursera & MathWorks:** Exploratory Data Analysis with MATLAB Jul 2020